

AI Summer School 2026
Medical Imaging Informatics

University of Pittsburgh

U-Net for Medical Image Segmentation

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Learning Objectives

After completing this lecture, you should be able to:

- Understand the U-Net architecture for medical image segmentation
- Explain encoder–decoder networks and skip connections
- Understand residual learning (ResNet)
- Recognize knee MRI and radiograph segmentation applications

Outline

- U-Net
- ResNet
- Hands-on Practice

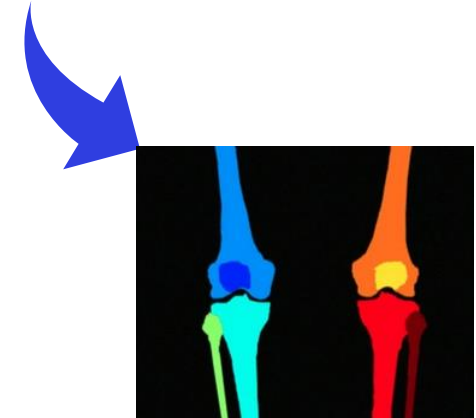
Why Segmentation in Medical Imaging?

■ Clinical Relevance

- Precise anatomical boundary delineation
- Quantitative analysis (volume, thickness, area measurements)
- Disease localization and characterization
- Treatment planning (e.g., surgery, radiotherapy)
- Longitudinal monitoring

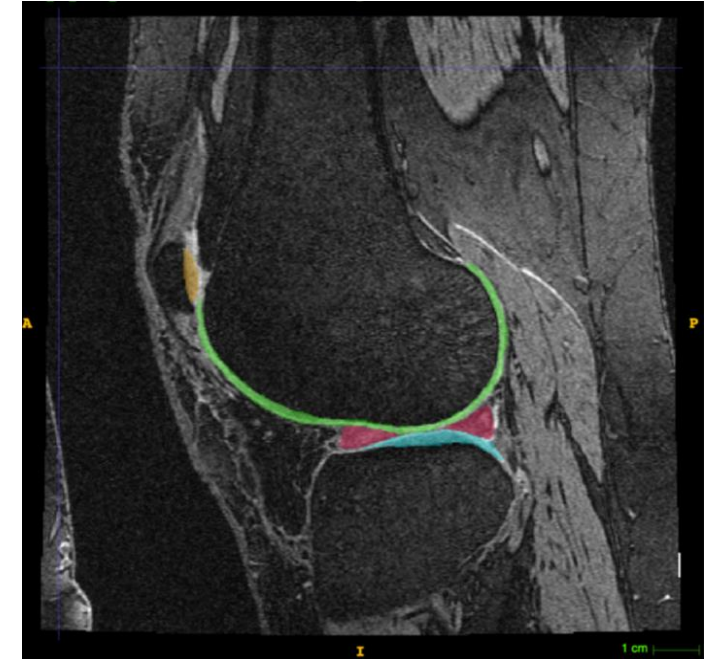
■ Technical & Informatics Perspective

- Enables pixel-level understanding
- Reduces inter-observer variability
- Supports automated clinical workflows
- Foundation for CAD systems



U-Net Motivation

- Limited availability of annotated medical imaging datasets
- Need for precise segmentation with contextual understanding
- Conventional CNNs suffer from spatial resolution loss
- Requirement for end-to-end, fully convolutional segmentation frameworks
- Designed specifically for biomedical image segmentation

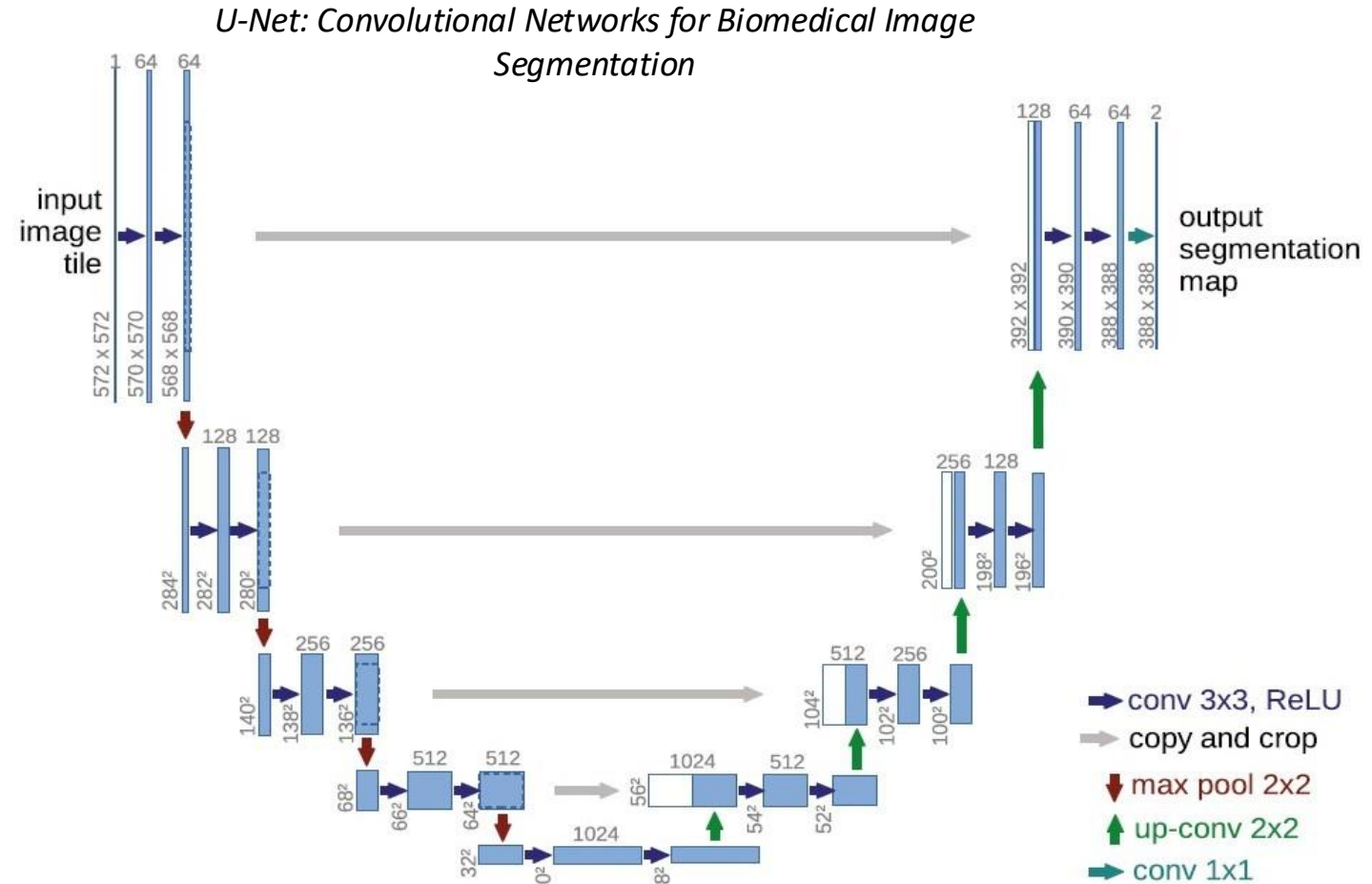


Manual expert annotation
(time-consuming & label-dependent)

U-Net Architecture

- Symmetric encoder–decoder architecture
- Contracting path (downsampling via convolution + max pooling)
- Expanding path (upsampling via transposed convolution)
- Skip connections for feature map concatenation

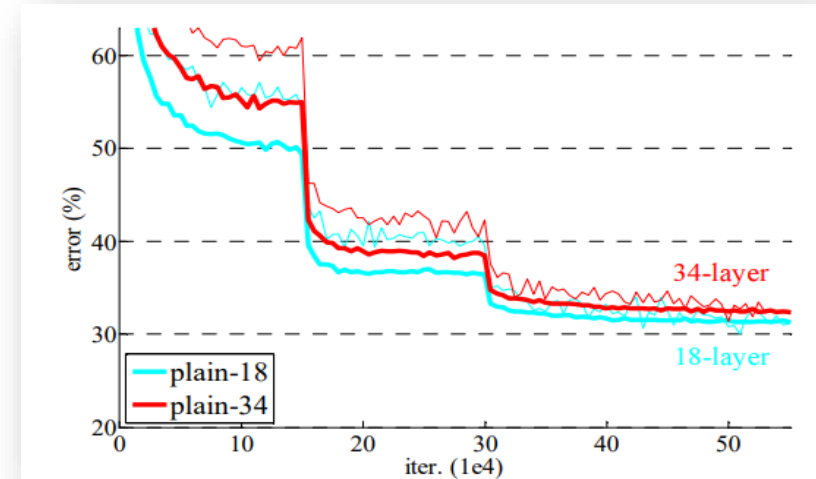
Fully convolutional design enabling pixel-wise segmentation



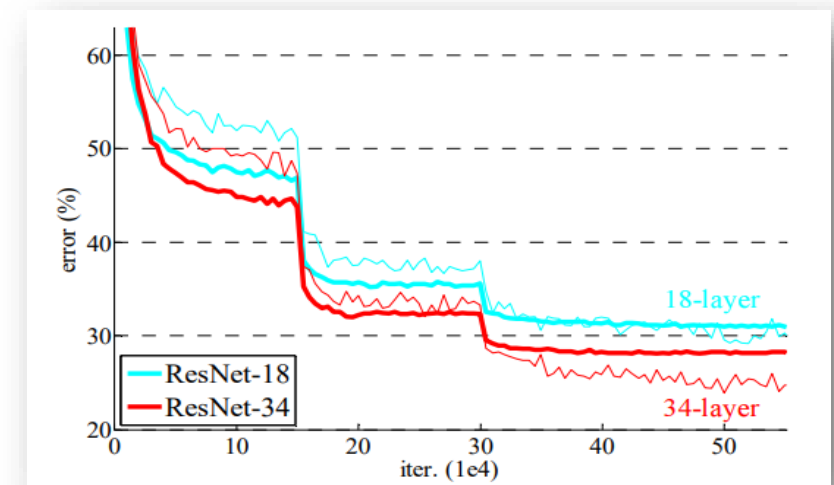
[Ronneberger et al., MICCAI 2015](#)

ResNet Motivation

- Deep networks promise better representations.
 - However, increasing depth does not guarantee better optimization.
- Vanishing Gradient
 - Gradient signals diminish in very deep networks.
- Degradation Problem
 - Deeper networks show higher training error, even without overfitting.
- Key Idea: Learn the residual function instead of the full mapping.
- Residual learning enables training of very deep backbones used in segmentation networks.



Deeper plain networks are harder to optimize



Deep Residual Learning for Image Recognition

[Kaiming He et al., 2016](#)

ResNet Architecture

■ Residual Block

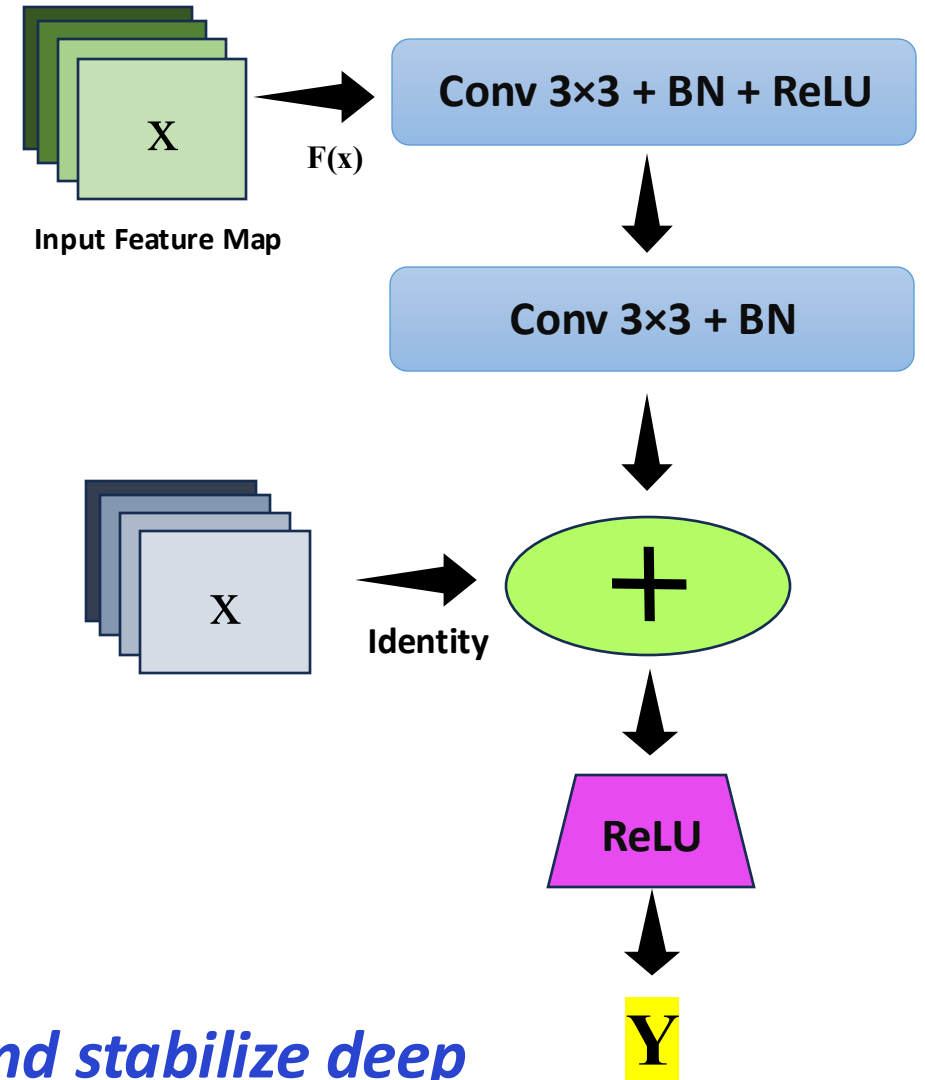
- Residual function: $F(x)$
- Identity shortcut connection
- Output formulation:

$$y = F(x) + x$$

Element-wise addition between input and residual mapping.

■ Core Design

- Enables stable optimization of very deep networks (50, 101, 152 layers)
- Batch Normalization
- ReLU activation



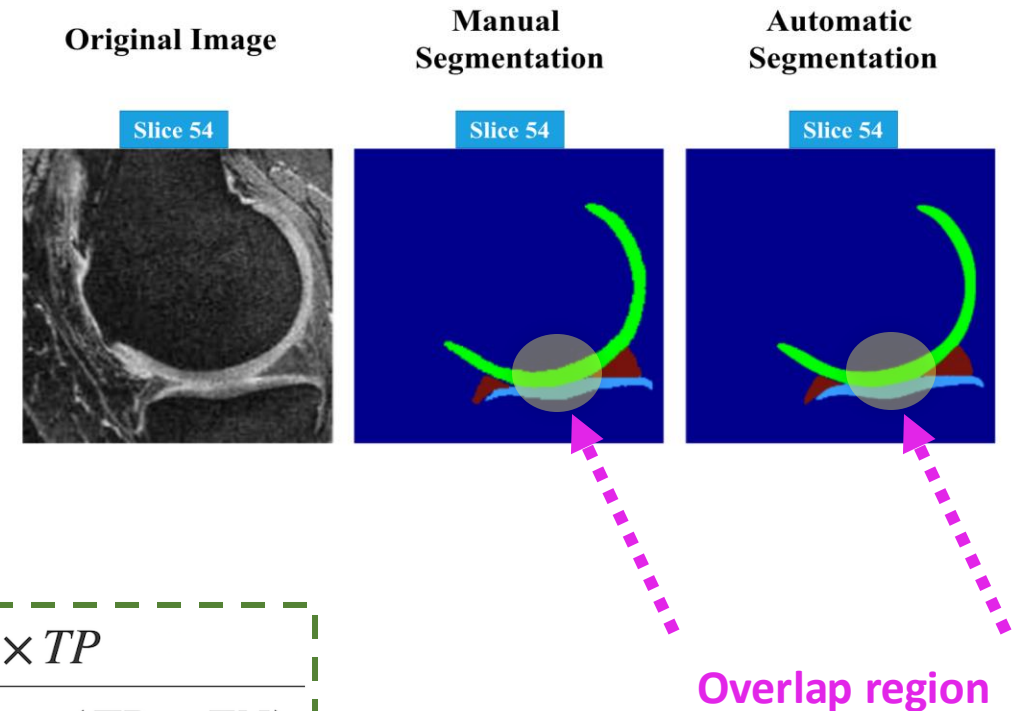
Residual connections facilitate gradient flow and stabilize deep optimization.

[Kaiming He et al., 2016](#)

Loss Functions

- **Pixel-wise Cross Entropy**
 - Computes loss for each pixel independently
 - Penalizes incorrect pixel predictions
- **Dice Loss**
 - Measures overlap between prediction and ground truth
 - Effective for imbalanced medical datasets
- **Combined Loss**
 - Combination of Cross Entropy and Dice Loss
 - Improves convergence and boundary accuracy

$$Dice = \frac{2 \times TP}{(TP + FP) + (TP + FN)}$$



Knee Joint Space Segmentation

- Automatic multi-class MRI segmentation
- U-Net + ResNet backbone
- Cross-Entropy + Dice loss
- Evaluated with IoU & Dice
- Uncertainty visualization (entropy maps)



KneeXNet-2.5D: an AI Tool set for Knee Cartilage and Meniscus Segmentation in MRIs

Upload either:

- A `.npy` file containing MRI slices, OR
- A folder (multiple files) of `.dcm` DICOM images

Upload NPY or DICOM files



Drag and drop files here
Limit 200MB per file • NPY, DCM

Browse files

Please upload a file or a folder of DICOM images.

Thank you!

Questions!



GitHub



Pitt AI Summer School